

IoT-enabled energy data management for EU ecodesign and energy labelling compliance in household washing machines, washer-dryers, and dishwashers manufacturing.

Manufacturing × Energy optimisation × IoT platform · OS049 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
68 is the world moving	69 can we play, can we win	HIGH 0 named in the evidence	€2.9bn observed evidence · per year	NOW budgeted procurement within 1...	L0 17 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying an IoT-enabled energy data management platform to help manufacturers of household washing machines, washer-dryers, and dishwashers comply with new EU ecodesign and energy labelling regulations. The platform collects and analyzes energy consumption data from production lines and products, enabling manufacturers to optimize energy use and demonstrate compliance. The need is immediate as the EU has recently set ecodesign and labelling requirements for these appliances.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.0bn €715m – €21.4bn	€2.9bn €513m – €15.3bn	€99.8m €17.9m – €536m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€733k €733k – €2.0m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting IoT platform	30.0 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

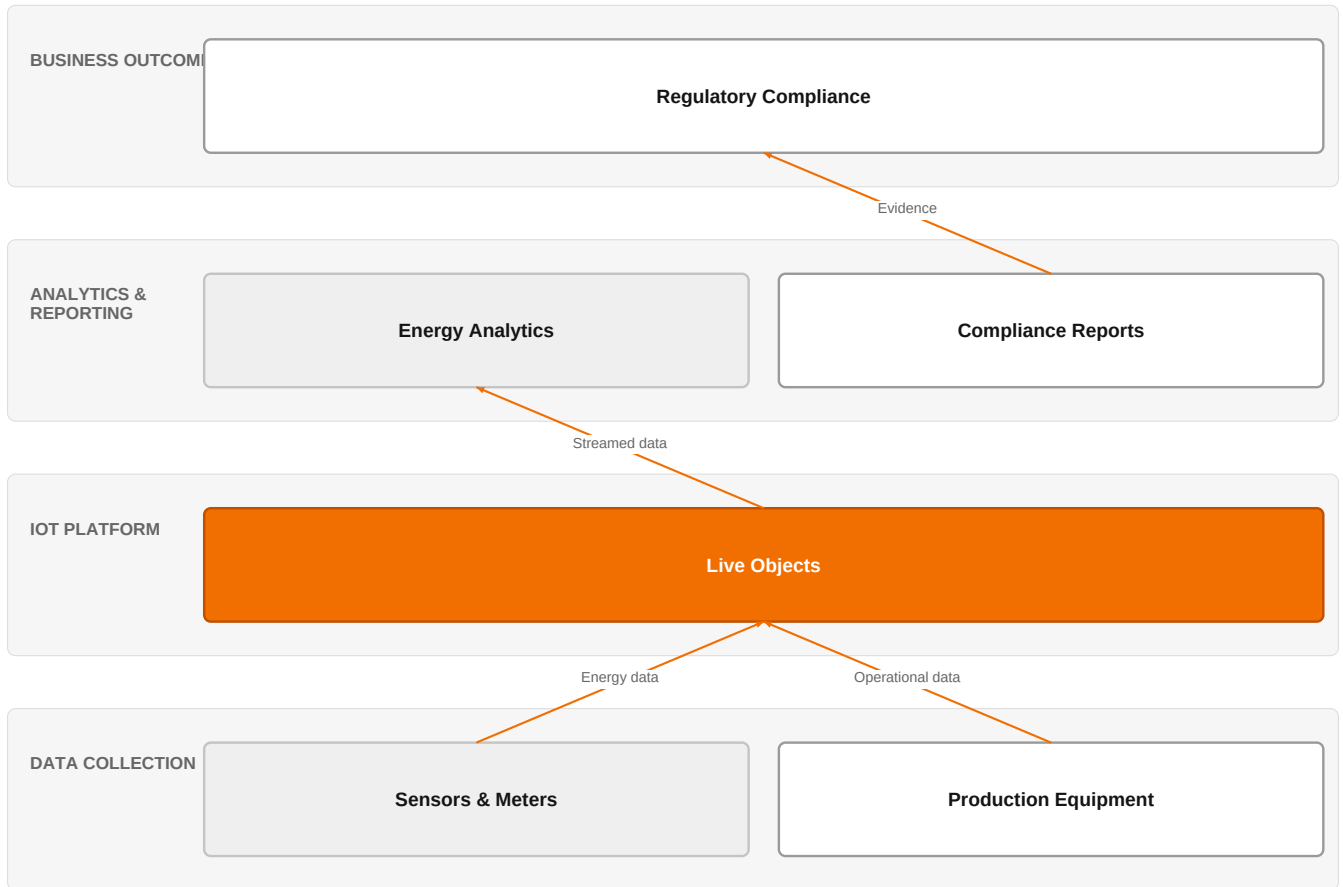
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

IoT Energy Compliance Platform



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's Live Objects platform collects energy data from production and sensors, enabling analytics and compliance reporting for EU regulations.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU has recently adopted ecodesign requirements for household washing machines, washer-dryers, and dishwashers, mandating energy efficiency improvements. Energy labelling requirements accompany these regulations, requiring clear communication of energy performance. These regulations create a compliance burden for manufacturers, who must now track and report energy data across their products and production processes. Additionally, there is a broader trend towards intelligent manufacturing, as seen in a traditional turbine factory transforming into an intelligent one, and research on digital twins for thermal energy management in manufacturing.

Sources: SIG-037778968624 ec.europa.eu 2026-05-05; SIG-CB4D5228C6C8 ec.europa.eu 2026-04-27; SIG-DB3990C67390 ec.europa.eu 2026-04-27; SIG-8F0739E8B909 ec.europa.eu 2026-04-23; SIG-14947CE171FF stdaily.com 2026-08-14; SIG-216688A4B953 openalex.org 2026-03-01

Who buys, and why now

The primary buyer is the head of manufacturing or operations at an appliance manufacturer. The pain is felt by compliance and sustainability teams who must ensure products meet EU energy efficiency standards and provide accurate labelling data. The trigger is the upcoming enforcement of the new ecodesign and labelling regulations, which forces manufacturers to invest in systems to measure, report, and optimize energy consumption. Without such systems, they risk non-compliance penalties and market access issues.

Why it is hot now — every claim bound to a dated source

- The EU has set ecodesign requirements for household washing machines, washer-dryers, and dishwashers, mandating energy efficiency improvements. [SIG-037778968624, SIG-CB4D5228C6C8]

- Energy labelling requirements accompany the ecodesign regulations, requiring clear communication of energy performance. [SIG-8F0739E8B909, SIG-DB3990C67390]
- EU ecodesign requirements mandate energy efficiency for household washing machines, washer-dryers, and dishwashers. [SIG-037778968624, SIG-CB4D5228C6C8]
- A traditional turbine factory has been transformed into an intelligent one, showing the potential for smart manufacturing. [SIG-14947CE171FF]
- The EU has set ecodesign requirements for household washing machines, washer-dryers, and dishwashers, mandating compliance for manufacturers. [SIG-037778968624, SIG-CB4D5228C6C8]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver an IoT-based energy data management platform built on Live Objects, Orange's IoT platform. The engagement would start with a discovery phase using Orange IoT experts to assess the manufacturer's current energy monitoring capabilities. Then, Live Objects would be deployed to collect energy data from production equipment and finished products, with integration into the manufacturer's existing systems. Orange Group researchers could provide advanced analytics for energy optimization. The solution would be certified to ISO 27001 for security, and TISAX if the manufacturer is in the automotive supply chain, though that is less likely for appliance makers.

Why Orange can win

Orange can win because it offers a secure, certified IoT platform (Live Objects) backed by a team of IoT experts and researchers. The platform is already proven in industrial settings, as evidenced by references like Colgate-Palmolive and Mondelez International. Orange's ISO 27001 certification ensures data security, which is critical for compliance data. However, the assets are not uniquely differentiated; competitors also offer IoT platforms. Orange must leverage its expertise and references to stand out.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; active in Manufacturing; competes in OX: Smart Industries	structural	—

NTT DATA	Global systems integrator	sells IoT platform; competes in OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells IoT platform	structural	—
Vodafone Business	Telecom operator peer	sells IoT platform; active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently tracking energy consumption across your production lines and products?
- 2 What is your timeline for ensuring compliance with the new EU ecodesign and energy labelling regulations?
- 3 Do you have a centralized system for collecting and reporting energy data, or is it siloed?
- 4 Have you already invested in any IoT or energy management solutions?
- 5 Who in your organization is responsible for regulatory compliance and energy reporting?
- 6 What are your biggest challenges in meeting the new energy efficiency targets?

Objections you will meet

They will say	You can answer
We already have an energy management system in place.	That's good. Our solution can complement your existing system by providing more granular, IoT-based data collection and analytics, helping you meet the new regulatory requirements more efficiently.
The regulations are not yet enforced, so we can wait.	True, but preparing now gives you a competitive advantage and avoids a last-minute rush. Early adoption can also help you optimize energy costs immediately.
We are considering a cheaper, simpler solution.	Cost is important, but our platform is designed for scalability and security, which is critical for compliance data. We also bring proven industrial expertise from references like Colgate-Palmolive.

Risks and unknowns

A key risk is that the new EU regulations may not yet be fully enforced, leading to delayed buying decisions. There is also uncertainty about the specific data requirements and whether manufacturers will need to invest in new systems or can adapt existing ones. Additionally, the competitive landscape is intense, with many established IoT platforms, so differentiation is challenging. Finally, the success of the solution depends on the manufacturer's willingness to share data and integrate with Orange's platform.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European household appliance manufacturer, outlining how Live Objects, combined with Orange's IoT experts and ISO 27001 certification, can deliver a compliant energy data management solution, referencing the smart factory transformation and the success at Colgate-Palmolive and Saint-Gobain Glass.
Sales	In a meeting with a European household appliance manufacturer's head of operations, say: 'Our Live Objects IoT platform, already used by companies like Colgate-Palmolive and Saint-Gobain Glass, can help you streamline energy data collection and reporting to meet EU ecodesign and energy labelling requirements.'
Strategist / Innovator	Commission a deep-dive brief on how Orange's Live Objects IoT platform and IoT expert pool can be positioned to support EU ecodesign and energy labelling compliance for household appliance manufacturers, using the Colgate-Palmolive, Mondelez International, Saint-Gobain Glass, Skytale, and Everise references as proof points of IoT-enabled energy management in manufacturing.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FC4FCB6FF0CC	2026-08-05	ted.europa.eu	1	Italy – Energy-management services – Gara europea per l'affidamento in concessione, tramite fin...
SIG-98F9B9FD3AF6	2026-07-22	ted.europa.eu	1	Portugal – Energy and related services – Contrato de Concessao (concecao, construcao, manutenca...
SIG-1C4F366EE4AA	2026-07-09	ted.europa.eu	1	Germany – Environmental indicators analysis other than for construction – One Health in der Pra...
SIG-ED6B4C16F384	2026-07-08	ted.europa.eu	1	Netherlands – Heat-exchange units, air-conditioning and refrigerating equipment, and filtering ...
SIG-AC86A5D91D50	2026-06-11	ted.europa.eu	1	France – Pollution tracking and monitoring and rehabilitation – SPCH a Harbonnieres (80) - Etud...
SIG-904EDD8843E8	2026-06-10	ted.europa.eu	1	Hungary – Energy-management services – Naperomu es tarolo villamos energia értékesítése
SIG-62C1F49E1CA5	2026-05-27	ted.europa.eu	1	Belgium – Heating engineering services for buildings – Raamovereenkomst: Externe expertise warm...
SIG-07C24AAAC8C8	2026-05-20	ted.europa.eu	1	Ireland – Energy and related services – Research and Policy Insights Framework
SIG-037778968624	2026-05-05	ec.europa.eu	1	Commission delegated act setting ecodesign requirements for household washing machines and hous...
SIG-CB4D5228C6C8	2026-04-27	ec.europa.eu	1	Setting ecodesign requirements for household dishwashers
SIG-DB3990C67390	2026-04-27	ec.europa.eu	1	Energy labelling requirements for household washing machines and household washer-dryers
SIG-8F0739E8B909	2026-04-23	ec.europa.eu	1	Energy labelling requirements for household dishwashers
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-216688A4B953	2026-03-01	openalex.org	1	Digital Twin-Enabled Thermal Energy Management System for Sustainable Manufacturing Process Opt...
SIG-955B81B681F1	2025-11-01	cordis.europa.eu	1	INTELLIGENT DIGITAL TOOLS FOR GREEN RENOVATION, ASSESSMENT, TRANSFORMATION, EFFICIENCY AND SUST...
SIG-14947CE171FF	2026-08-14	stdaily.com	2	Traditional Turbine Factory Turns Intelligent
SIG-288B4B508030	2026-08-10	arxiv.org	3	Automated generation of experimentally validated digital twins for desiccant-based low-dew-poin...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:23+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.